

Project 1: TeamForge — Software Project Management & Team Collaboration Platform

2. Problem Statement – What problem does the project aim to solve?

Software development teams often use multiple disconnected tools for managing tasks, communication, deadlines, and project progress. This makes it difficult to know who is responsible for what, which tasks are pending, and how much of the project has been completed. **TeamForge aims to provide a centralized platform for managing software projects, teams, tasks, and development workflows.**

3. Target Users – Who will use the application?

- Project Managers
- Team Leaders
- Software Developers / Team Members
- Organization Administrators

4. Project Objectives – What do you want to achieve?

- Centralize software project and task management.
- Make team responsibilities and project progress easier to track.
- Provide a structured workflow for software development tasks.
- Support Agile concepts such as Kanban and sprint management.
- Improve collaboration between team members.
- Provide project managers with an overview of project progress.

5. Major Features / Functionalities – What will the system be able to do?

- **User & Team Management** — Manage users, teams, roles, and permissions.
- **Project Management** — Create projects, add members, set deadlines, and track progress.
- **Task & Issue Management** — Create, assign, prioritize, and track tasks and issues.
- **Kanban Board** — Move tasks through To Do, In Progress, Review, Testing, and Done.
- **Sprint Management** — Create sprints, select tasks, define goals, and track sprint progress.
- **Team Collaboration** — Add comments and share project-related files.
- **Project Dashboard** — View completed, pending, and overdue tasks and overall progress.
- **Activity & Notifications** — Track important activities and notify users about assignments and deadlines.

6. Proposed Technology Stack – Frontend, backend, database, frameworks/tools, etc.

- **Frontend:** React.js, Tailwind CSS
- **Backend:** Node.js, Express.js
- **Database:** PostgreSQL
- **ORM:** Prisma
- **Authentication:** JWT, bcrypt
- **Testing/API:** Postman, Jest/Supertest
- **Tools:** Git, GitHub, Docker

7. Basic System Workflow – How will the application work from a user's perspective?

User registers/logs in → joins or creates a team → Project Manager creates a project → team members are added → tasks are created and assigned → members work on tasks → tasks move through the Kanban workflow → members collaborate through comments → Project Manager monitors progress through the dashboard → project is completed.

8. Uniqueness / Significance – What makes the project useful, interesting, or different?

TeamForge focuses specifically on the **software development workflow** rather than being a simple task-management system. It combines team management, project management, task tracking, Kanban workflow, sprint planning, collaboration, and progress monitoring in one platform. It can be useful for student software projects as well as small development teams.

9. Feasibility and Project Scope – Can it realistically be completed within the course duration?

Yes. The project can be developed incrementally.

Core scope:

- Authentication & RBAC
- Team management
- Project management
- Task management
- Kanban board
- Comments
- Basic dashboard

Optional extensions:

- Sprint management

- Task dependencies
- Notifications
- Activity logs
- Advanced analytics

The core version is realistic for a course project, while additional features can be added if time permits.

10. Expected Outcome – What will the final application provide?

A functional web-based platform where software teams can **create projects, organize members, assign and manage tasks, track development through a Kanban workflow, collaborate with teammates, and monitor overall project progress from one centralized system.**

Project 2: BatchHub — Centralized Academic Management & Collaboration Platform

2. Problem Statement – What problem does the project aim to solve?

Academic information for university students is often scattered across Messenger/WhatsApp groups, Facebook groups, Google Drive, PDFs, and individual conversations. Students may miss important notices, struggle to find course materials, forget assignment deadlines, or have difficulty finding organized project-group information. **BatchHub aims to centralize important academic information and collaboration tools in one platform.**

3. Target Users – Who will use the application?

- University Students
- Class Representatives / Batch Moderators
- Teachers / Course Instructors
- Department or System Administrators

4. Project Objectives – What do you want to achieve?

- Centralize academic information for a batch or department.
- Make notices, notes, assignments, and resources easier to access.
- Help students track academic deadlines.
- Provide an organized class routine.
- Support academic project groups and collaboration.
- Reduce dependence on scattered communication channels.

- Provide administrators with control over academic content.

5. Major Features / Functionalities – What will the system be able to do?

- **Notices** — Publish and organize class, exam, department, and other academic announcements.
- **Class Notes & Resources** — Upload, search, categorize, and download notes, slides, PDFs, previous questions, and lab materials.
- **Assignments** — Publish assignment details, deadlines, instructions, and related files.
- **Class Routine** — Display weekly classes with subjects, teachers, rooms, and schedules.
- **Academic Project Groups** — Create groups, manage members, assign project tasks, and share project files.
- **Student Profiles** — Maintain student information and optionally display GitHub/LinkedIn profiles.
- **Search & Filtering** — Search academic resources, notices, assignments, and other information.
- **Notifications** — Notify students about new notices, assignments, deadlines, and academic updates.
- **Admin/Moderator Panel** — Manage users and moderate uploaded academic content.

6. Proposed Technology Stack – Frontend, backend, database, frameworks/tools, etc.

- **Frontend:** React.js, Tailwind CSS
- **Backend:** Node.js, Express.js
- **Database:** PostgreSQL
- **ORM:** Prisma
- **Authentication:** JWT, bcrypt
- **File Storage:** Cloudinary / similar storage service
- **Testing/API:** Postman, Jest/Supertest
- **Tools:** Git, GitHub, Docker

7. Basic System Workflow – How will the application work from a user's perspective?

Student registers/logs in → joins their batch/course → accesses the dashboard → views notices, routine, assignments, and resources → searches or downloads academic materials → tracks assignment deadlines → joins or creates academic project groups → collaborates with group members → receives notifications about important academic updates.

8. Uniqueness / Significance – What makes the project useful, interesting, or different?

BatchHub combines several academic activities that are normally scattered across different platforms into one centralized system. Instead of using separate communication groups, cloud folders, PDFs, and spreadsheets for notices, resources, assignments, routines, and project information, students can access them through a single platform designed specifically around their academic workflow.

9. Feasibility and Project Scope – Can it realistically be completed within the course duration?

Yes, if the project is developed in phases.

Core scope:

- Authentication & user management
- Notices
- Class routine
- Assignments
- Academic resources
- Student/batch management
- Basic dashboard

Optional extensions:

- Project-group management
- Resource moderation
- Notifications
- Assignment submission
- Bookmarks/ratings
- Advanced search

The core system is realistic for a course project, while the optional features can be added depending on available time.

10. Expected Outcome – What will the final application provide?

A centralized academic platform where students can **access notices, manage assignments, view class schedules, find and share academic resources, participate in project groups, and receive important academic updates**, while teachers and administrators can manage and organize academic information.

